08-25 第1篇

阅读理解（说明文） How Do Green Plants "Eat" Sunlight?

① Take a quiet walk in any park on a summer morning, and you will see thousands of green leaves moving gently in the breeze. It looks peaceful, but inside each leaf, a very busy factory is running. Plants do not have mouths, yet they grow taller, heavier and stronger every day. How? The answer is that green plants have a special way to "eat" — **not with their mouths, but with their leaves and the sun**. This ancient skill is called **photosynthesis** (a big word that simply means "making with light").

② Every green plant needs **three things** to run its sunlight factory:

- **Sunlight**, which is the energy;
- **Water**, which comes from the ground through the **roots**;
- **Carbon dioxide**, which is an invisible gas in the air (the same gas we breathe out).

③ The factory itself is not in the plant's stem or roots. It is inside each leaf, in tiny green "rooms" called **chloroplasts**. Every chloroplast is full of a green **chemical** called **chlorophyll**. This chlorophyll is what makes the leaf look green in the first place. Its real job is to **catch the energy of sunlight**, just as a small solar panel catches light on a roof.

④ Here is a simple picture of what happens inside each leaf:

1. **Sunlight lands on the leaf.** The chlorophyll inside the chloroplasts grabs the light energy.
2. **Water comes up from the roots** through small tubes, like a silent underground water pipe.
3. **Carbon dioxide enters the leaf** through tiny openings on the back of the leaf, called **stomata**.
4. Using the light energy, the leaf mixes water and carbon dioxide together and builds a simple kind of **sugar** called **glucose**. This sugar is the plant's real food.
5. **Oxygen** is left over from this process. The leaf releases the oxygen back into the air through the same tiny stomata.

⑤ This quiet, daily work is more important than many people realise. First, **the sugar made in leaves is the starting food of almost every living thing** on Earth. A cow that eats grass is eating the sugar made by leaves. A child who drinks milk is drinking something that began as sugar in grass. Even a pot of rice on our dinner table is, in the end, sunlight stored by small rice plants.

⑥ Second, the **oxygen** we breathe comes almost entirely from green plants and from tiny sea plants in the ocean. Every tree in a city park quietly offers oxygen every minute of the day. Large rainforests — especially those in the Amazon — are sometimes called "the lungs of the Earth" because they produce such huge amounts of oxygen and absorb so much carbon dioxide.

⑦ Third, photosynthesis helps to **control the Earth's air**. The more carbon dioxide plants take in, the less "warming gas" stays in our atmosphere. This is one important reason why scientists say that planting trees and protecting forests matter in the fight against climate change.

⑧ Of course, plants do not work all year round. When the weather becomes very cold or very dark, many trees drop their leaves and close their factories for the winter. But as soon as spring sunlight returns, new leaves open, chlorophyll turns green, and the silent song of photosynthesis begins again — the same song that has been sung on Earth for more than three billion years.

⑨ So next time you see a simple green leaf on a bus stop tree, remember that it is not just decoration. It is a tiny, ancient factory turning sunlight, air and water into the food and oxygen that feed the whole living world — including you.

注释： photosynthesis 光合作用；chloroplast 叶绿体；chlorophyll 叶绿素；glucose 葡萄糖；stomata 气孔

根据文章内容，选择最佳答案。

1. What three things do green plants need to do photosynthesis?

A. Sunlight, soil, and wind.
B. Sunlight, water and carbon dioxide.
C. Sunlight, oxygen and rain.
D. Sunlight, sugar and electricity.

2. Where exactly inside a plant does photosynthesis happen?

A. In the roots.
B. In tiny green rooms called chloroplasts, inside each leaf.
C. In the centre of the stem.
D. Only in the flower.

3. According to the passage, what is the real job of chlorophyll?

A. To make the leaf heavy.
B. To catch the energy of sunlight for the leaf, like a small solar panel.

C. To push the plant up in the air.
D. To carry water up from the roots.

4. Why are large rainforests sometimes called "the lungs of the Earth"?

A. Because they look like human lungs from the sky.
B. Because they breathe with two holes like a human nose.
C. Because they produce huge amounts of oxygen and absorb a lot of carbon dioxide.
D. Because they are full of wild lions.

5. What is the writer's main purpose in this passage?

A. To teach readers how to grow flowers in a pot.
B. To explain how plants turn sunlight, water and carbon dioxide into food and oxygen, and to help readers feel the quiet importance of green leaves.
C. To compare plants in different countries.
D. To prove that plants are smarter than animals.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "Sunlight ... Water ... Carbon dioxide"
【解析】原文明确列出阳光、水、二氧化碳三要素。A、C、D 均与原文不符。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "inside each leaf, in tiny green 'rooms' called chloroplasts"
【解析】原文直接给出位置。A、C、D 均与原文不符。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第③段 "Its real job is to catch the energy of sunlight, just as a small solar panel catches light on a roof"
【解析】原文使用比喻，叶绿素如同太阳能板。A、C、D 均与原文不符。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑥段 "Large rainforests ... produce such huge amounts of oxygen and absorb so much carbon dioxide"
【解析】雨林产氧吸碳故得此称。A、B、D 均与原文不符。故选 C。

5.
【答案】B
【题型】写作意图题
【定位】第①段引入 & 第⑨段结尾 "a tiny, ancient factory turning sunlight, air and water into the food and oxygen"
【解析】作者目的是解释光合作用并引发读者对绿叶的敬意。A、C、D 均与文意不符。故选 B。